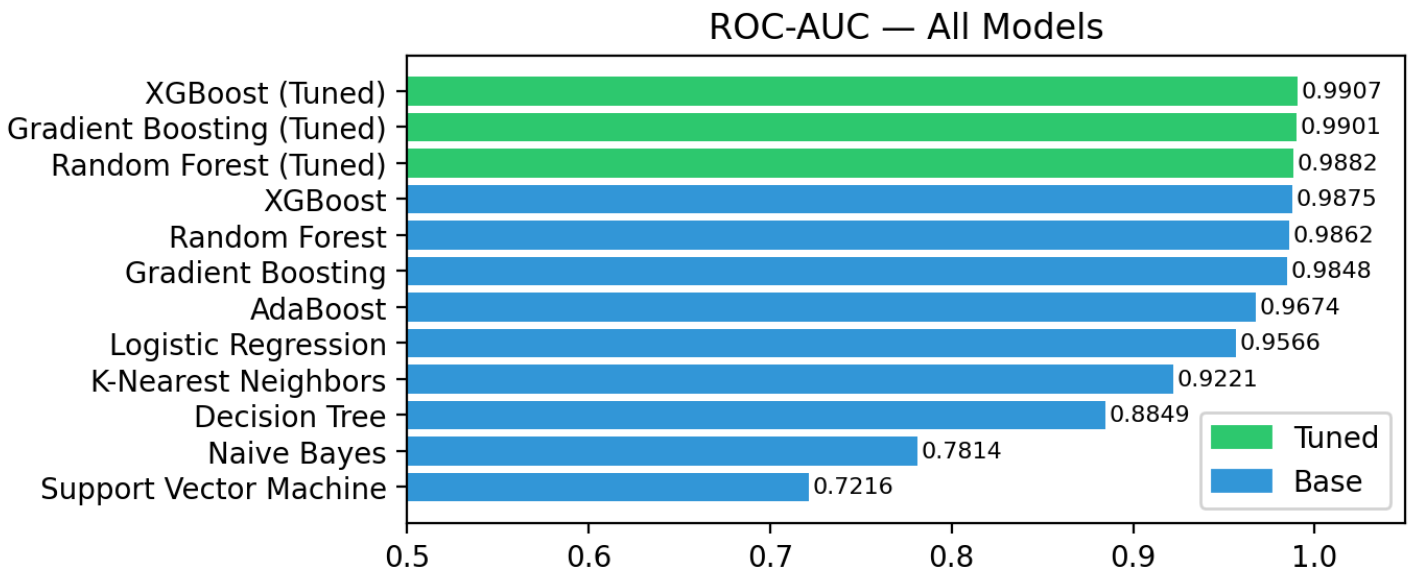
Model Leaderboard — All Models

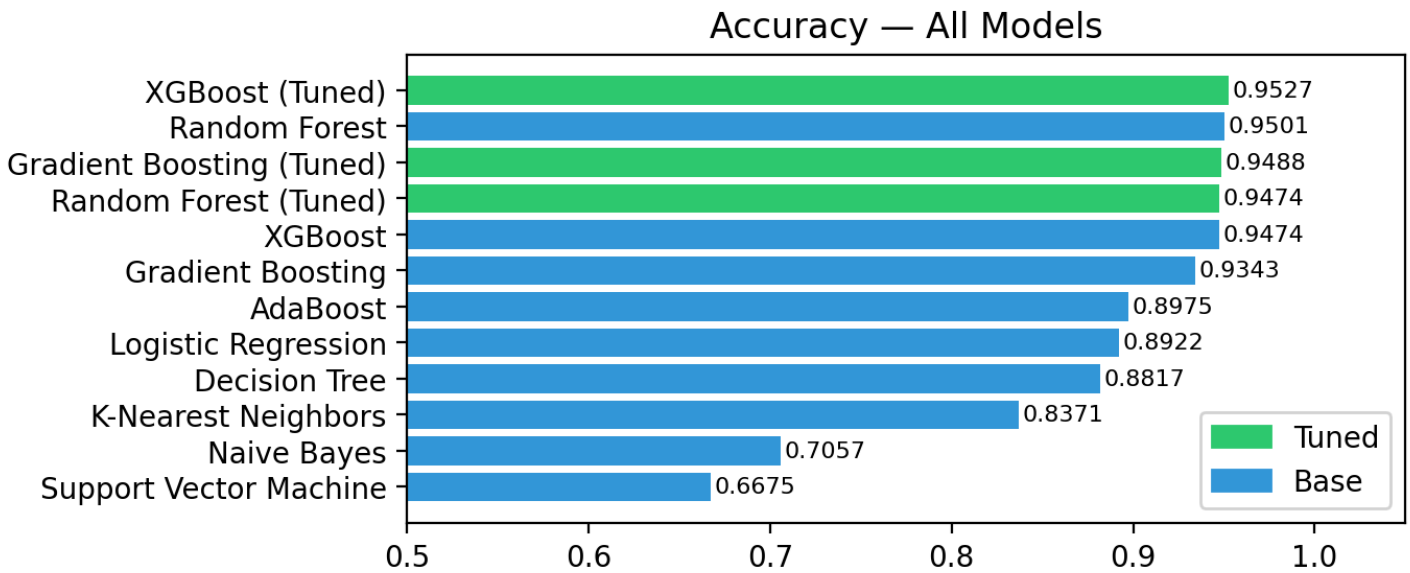
	Medal	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	🥇	**XGBoost (Tuned)**	**0.9527**	**0.9574**	**0.9474**	**0.9524**	**0.9907**
1	🥈	Gradient Boosting (Tuned)	0.9488	0.9523	0.9447	0.9485	0.9901
2	🥉	Random Forest (Tuned)	0.9474	0.9381	0.9579	0.9479	0.9882
3		XGBoost	0.9474	0.9570	0.9368	0.9468	0.9875
4		Random Forest	0.9501	0.9407	0.9605	0.9505	0.9862
5		Gradient Boosting	0.9343	0.9388	0.9289	0.9339	0.9848
6		AdaBoost	0.8975	0.8932	0.9026	0.8979	0.9674
7		Logistic Regression	0.8922	0.9139	0.8658	0.8892	0.9566
8		K-Nearest Neighbors	0.8371	0.7678	0.9658	0.8555	0.9221
9		Decision Tree	0.8817	0.8816	0.8816	0.8816	0.8849
10		Naive Bayes	0.7057	0.7321	0.6474	0.6872	0.7814

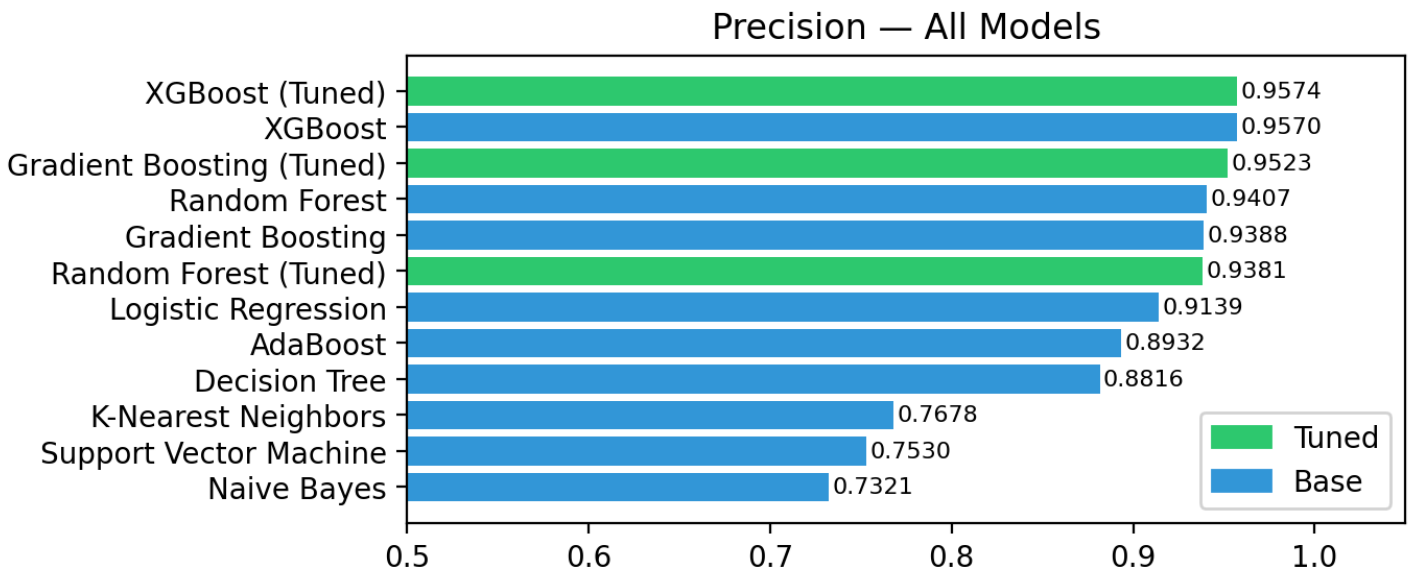
😄 Model Comparison Dashboard

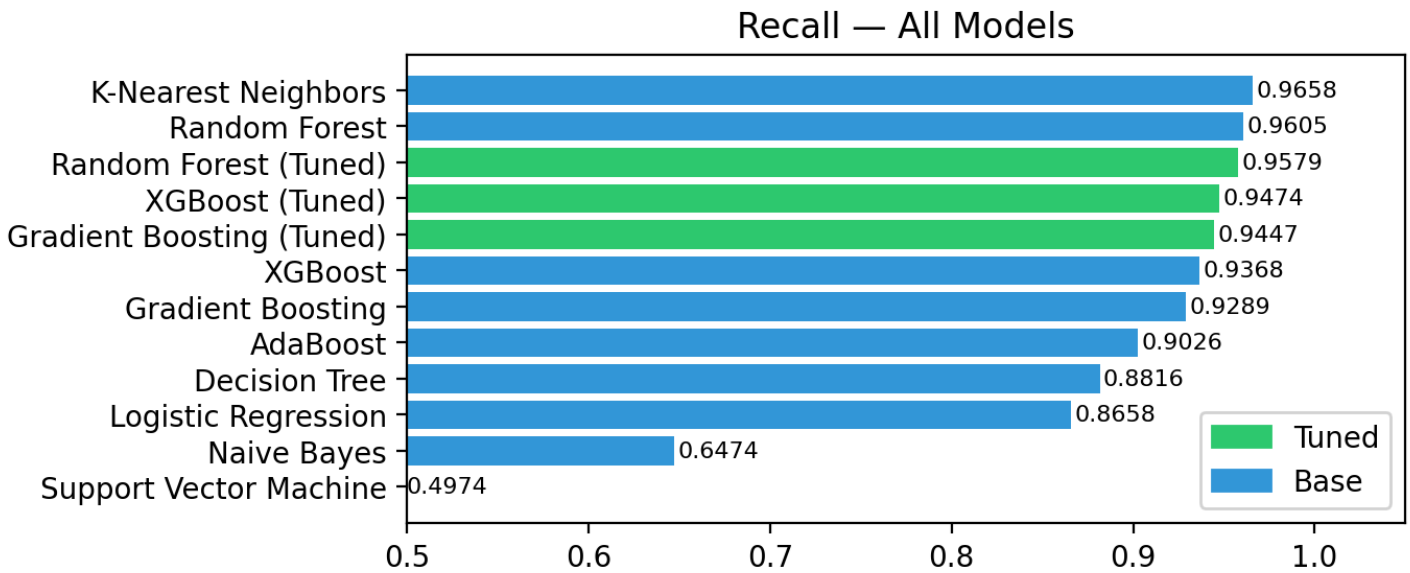
Customer Segmentation & Campaign Response Prediction

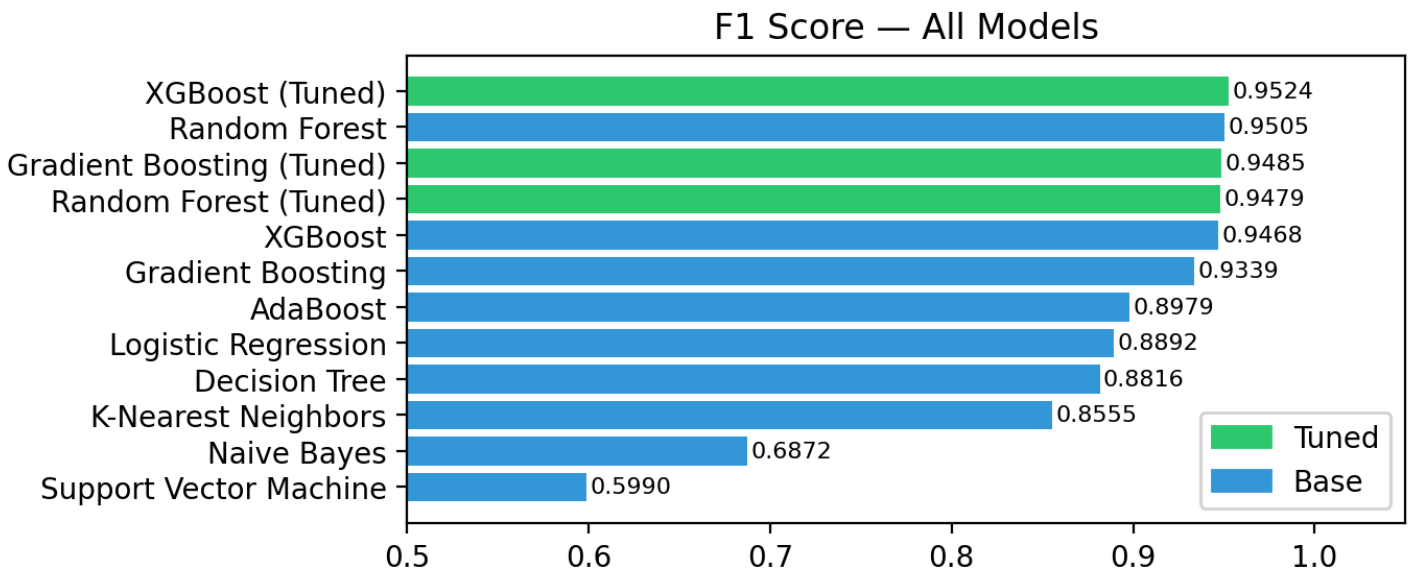
🎯 Accuracy	✅ Precision	🔍 Recall	⚖️ F1 Score	📈 ROC-AUC
95.27%	95.74%	94.74%	95.24%	99.07%

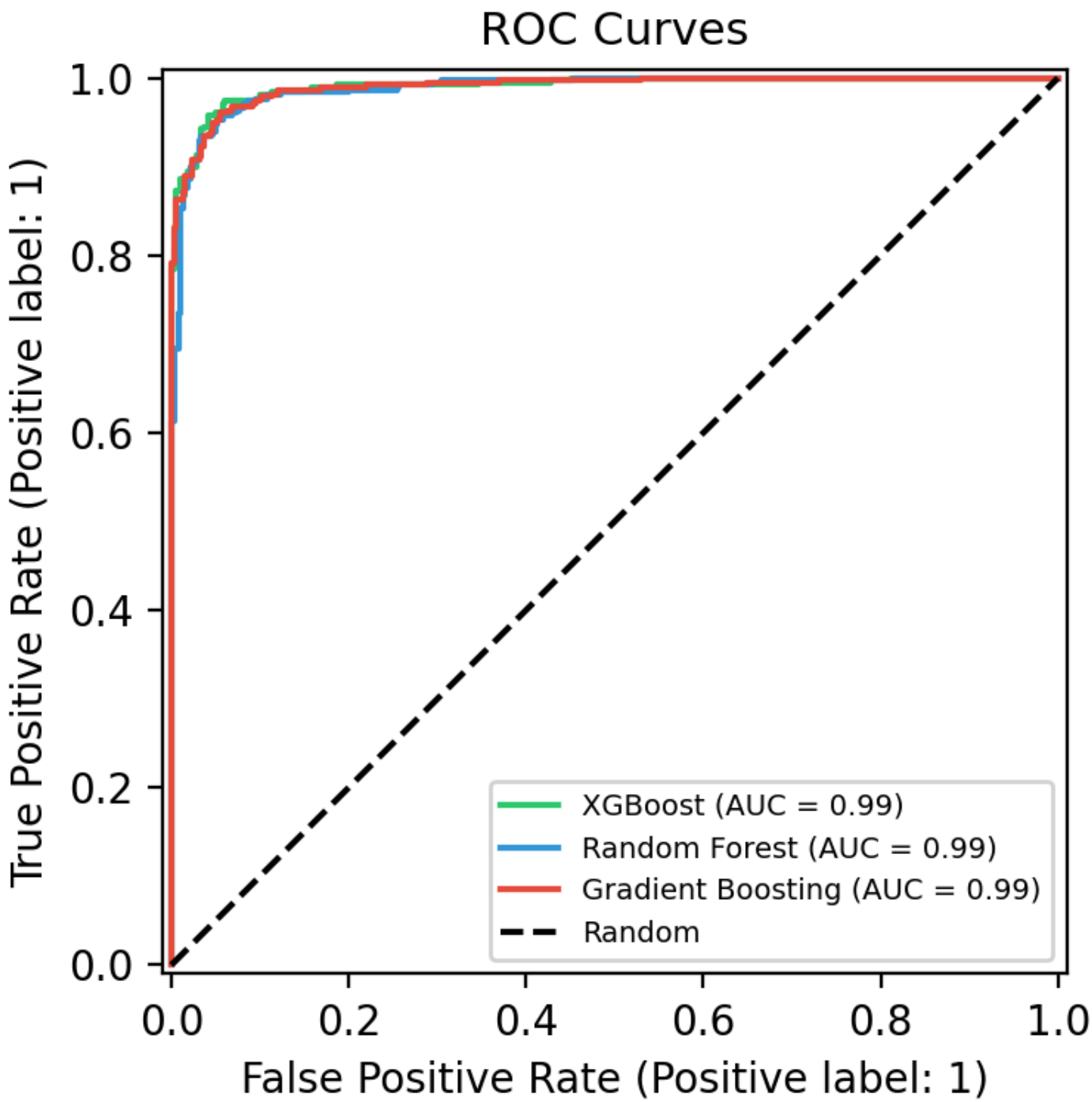

ROC-AUC — All Models
XGBoost (Tuned)
Gradient Boosting (Tuned)
Random Forest (Tuned)
XGBoost
Random Forest
Gradient Boosting
AdaBoost
Logistic Regression
K-Nearest Neighbors
Decision Tree
Naive Bayes
Support Vector Machine
0.9907
0.9901
0.9882
0.9875
0.9862
0.9848
0.9674
0.9566
0.9221
0.8849
0.7814
0.7216
Tuned
Base
0.5
0.6
0.7
0.8
0.9
1.0

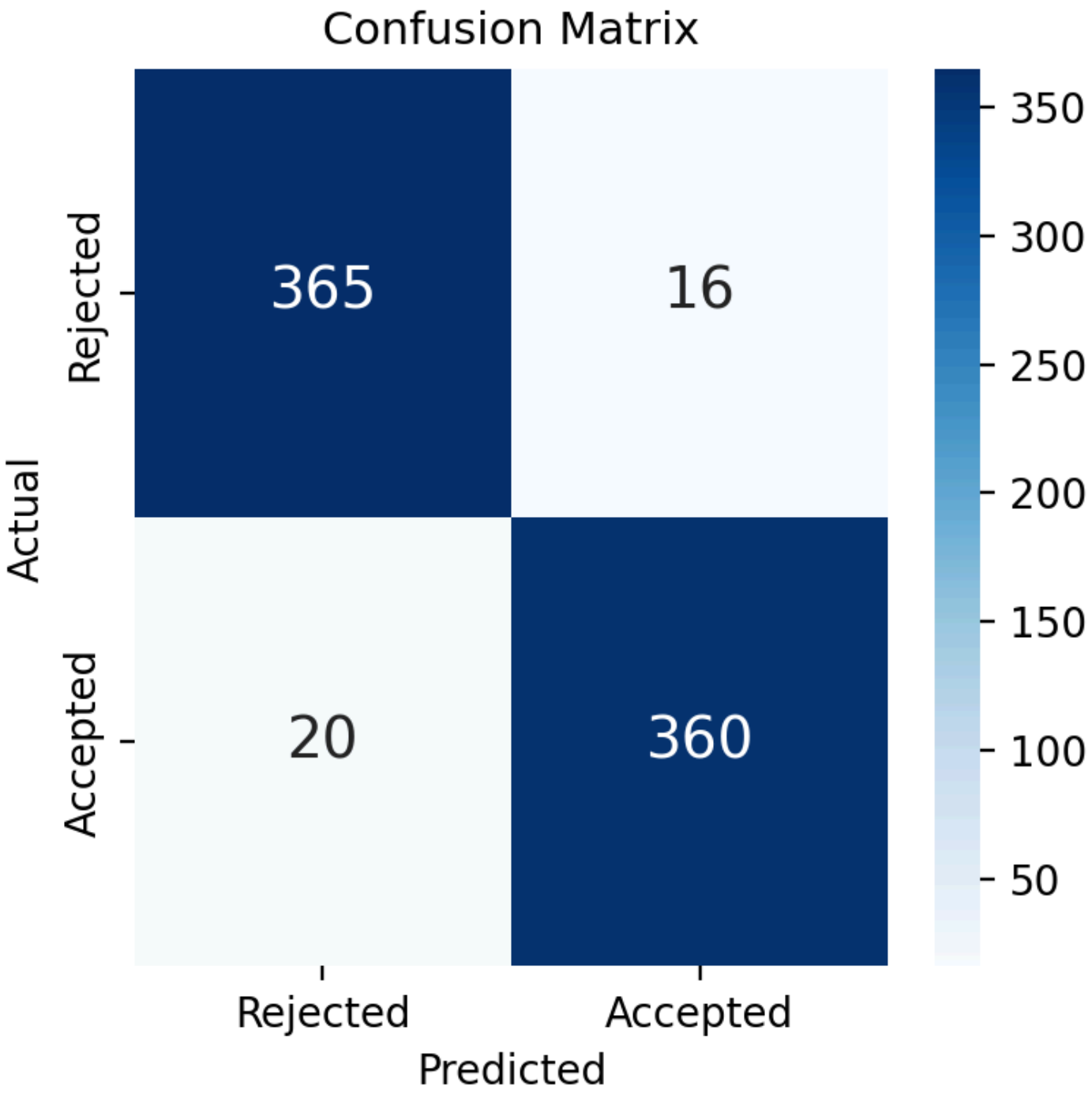

Accuracy — All Models
XGBoost (Tuned)
Random Forest
Gradient Boosting (Tuned)
Random Forest (Tuned)
XGBoost
Gradient Boosting
AdaBoost
Logistic Regression
Decision Tree
K-Nearest Neighbors
Naive Bayes
Support Vector Machine
0.9527
0.9501
0.9488
0.9474
0.9474
0.9343
0.8975
0.8922
0.8817
0.8371
0.7057
0.6675
Tuned
Base
0.5
0.6
0.7
0.8
0.9
1.0

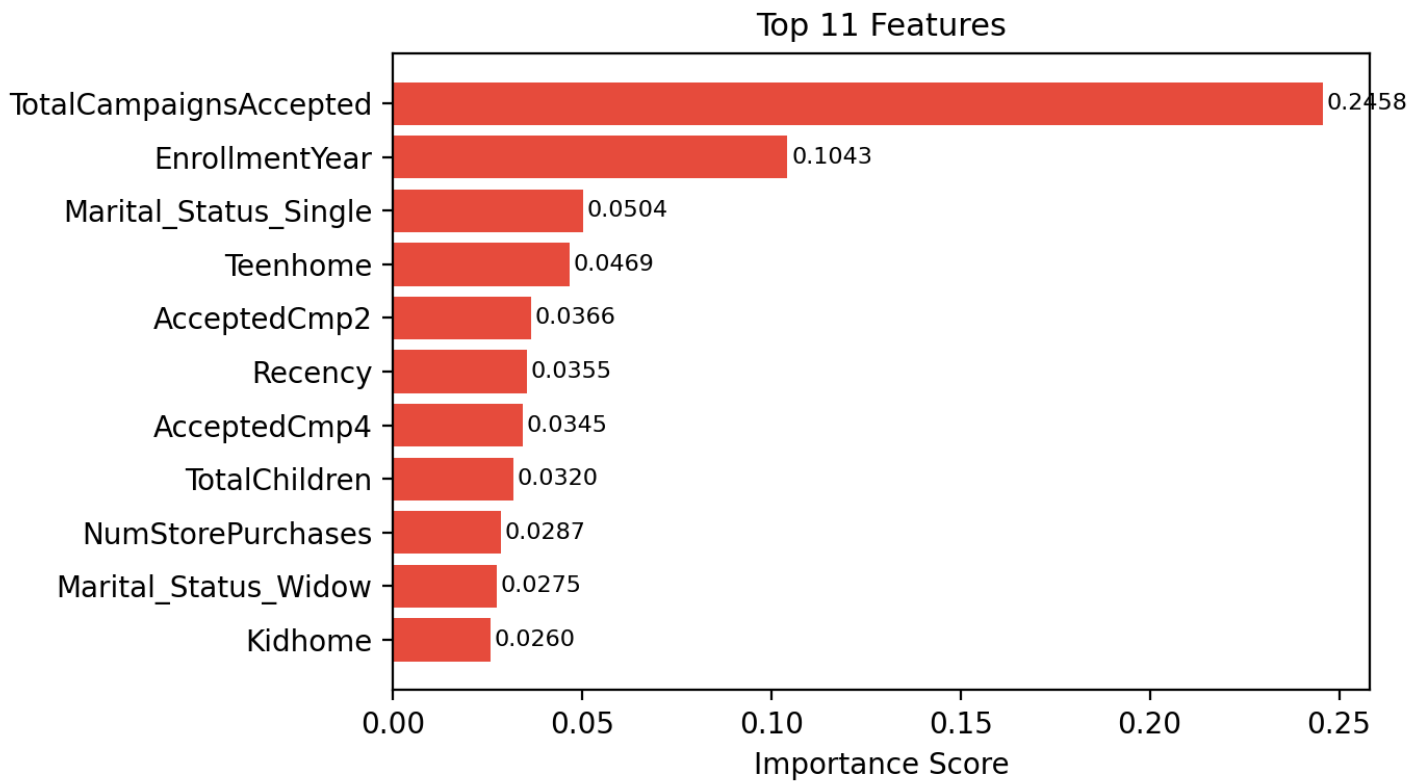

Precision — All Models
XGBoost (Tuned)
XGBoost
Gradient Boosting (Tuned)
Random Forest
Gradient Boosting
Random Forest (Tuned)
Logistic Regression
AdaBoost
Decision Tree
K-Nearest Neighbors
Support Vector Machine
Naive Bayes
0.9574
0.9570
0.9523
0.9407
0.9388
0.9381
0.9139
0.8932
0.8816
0.7678
0.7530
0.7321
Tuned
Base
0.5
0.6
0.7
0.8
0.9
1.0


Recall — All Models
K-Nearest Neighbors
Random Forest
Random Forest (Tuned)
XGBoost (Tuned)
Gradient Boosting (Tuned)
XGBoost
Gradient Boosting
AdaBoost
Decision Tree
Logistic Regression
Naive Bayes
Support Vector Machine
0.9658
0.9605
0.9579
0.9474
0.9447
0.9368
0.9289
0.9026
0.8816
0.8658
0.6474
0.4974
Tuned
Base
0.5
0.6
0.7
0.8
0.9
1.0


F1 Score — All Models
XGBoost (Tuned)
Random Forest
Gradient Boosting (Tuned)
Random Forest (Tuned)
XGBoost
Gradient Boosting
AdaBoost
Logistic Regression
Decision Tree
K-Nearest Neighbors
Naive Bayes
Support Vector Machine
0.9524
0.9505
0.9485
0.9479
0.9468
0.9339
0.8979
0.8892
0.8816
0.8555
0.6872
0.5990
Tuned
Base
0.5
0.6
0.7
0.8
0.9
1.0


ROC Curves
True Positive Rate (Positive label: 1)
False Positive Rate (Positive label: 1)
0.0
0.2
0.4
0.6
0.8
1.0
XGBoost (AUC = 0.99)
Random Forest (AUC = 0.99)
Gradient Boosting (AUC = 0.99)
Random


Confusion Matrix
365
16
20
360
Rejected
Accepted
Actual
Rejected
Accepted
Predicted
350
300
250
200
150
100
50



Business Insights Dashboard

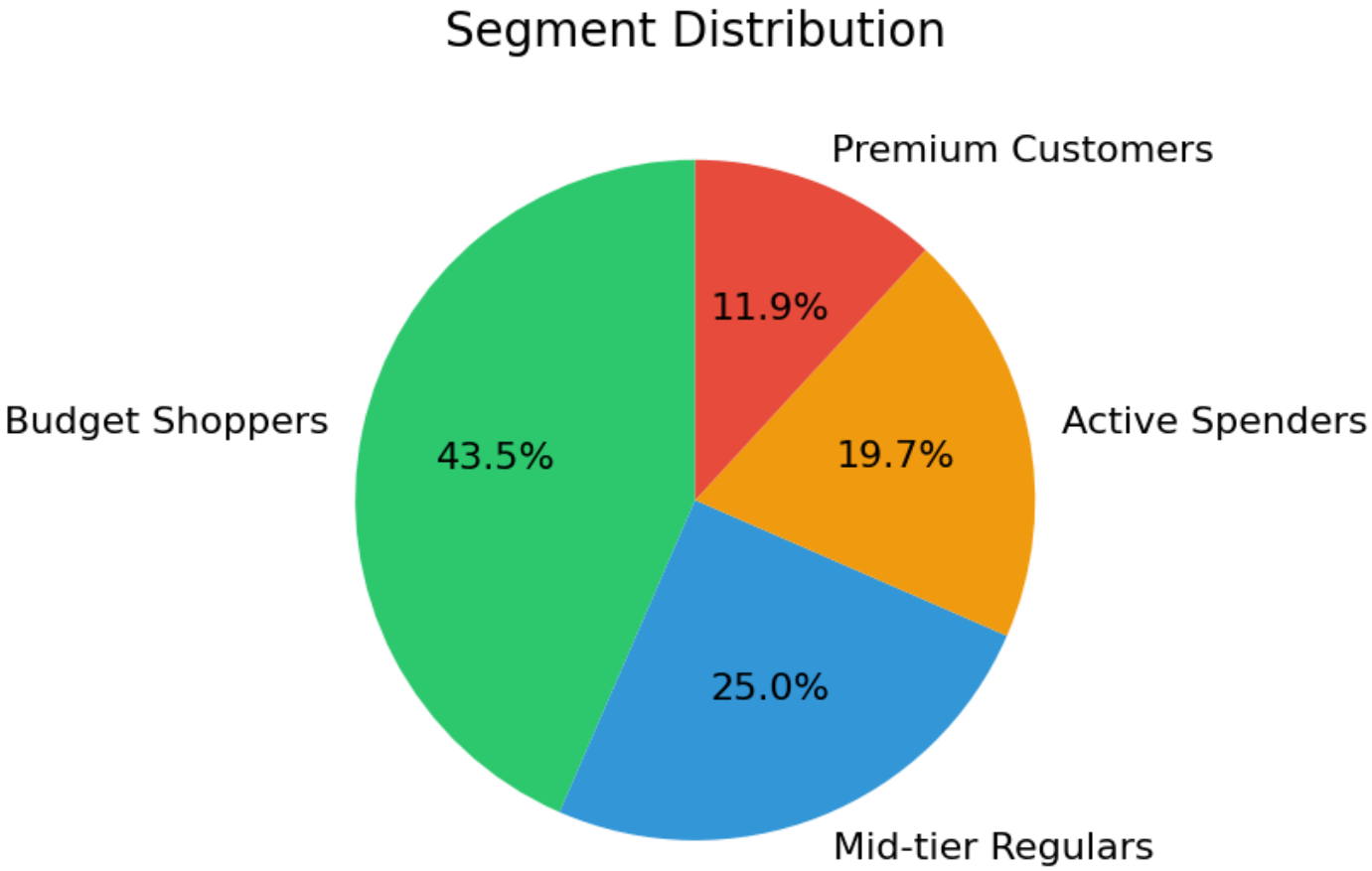
Customer Segmentation & Marketing Intelligence

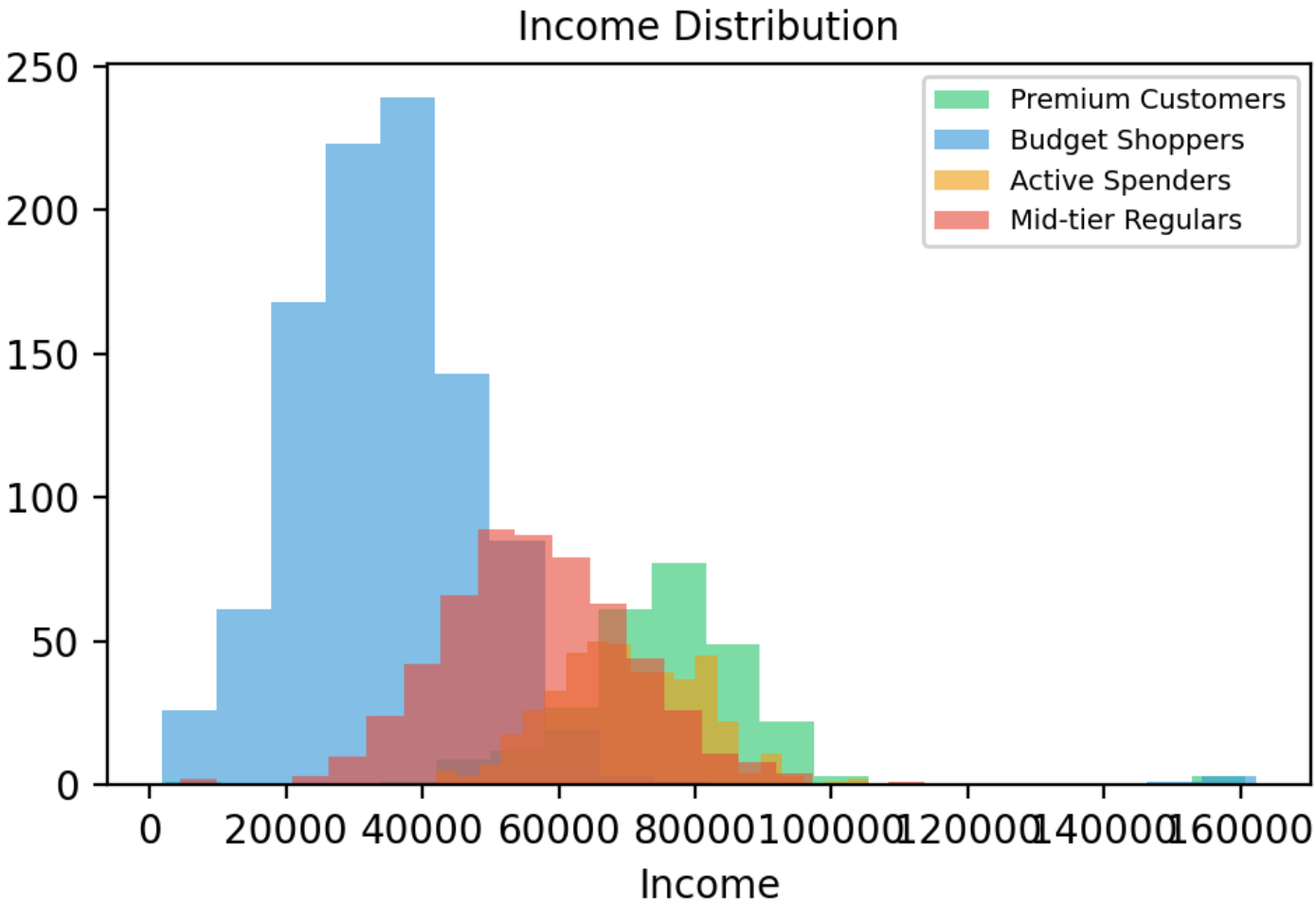
Total Customers	Campaign Accepted	Acceptance Rate	Avg Income
2,236	334	14.9%	$51,953

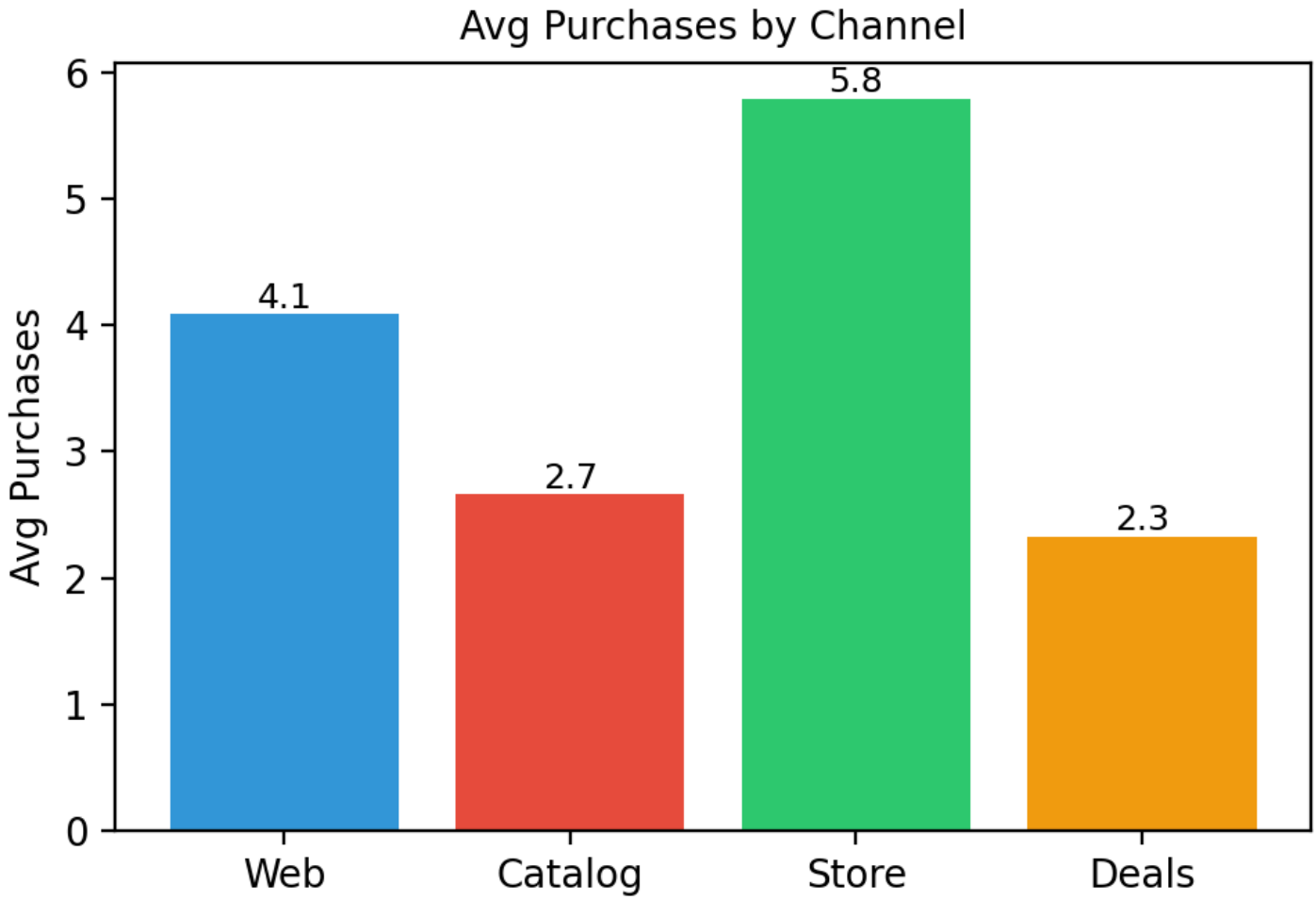
Segment Overview

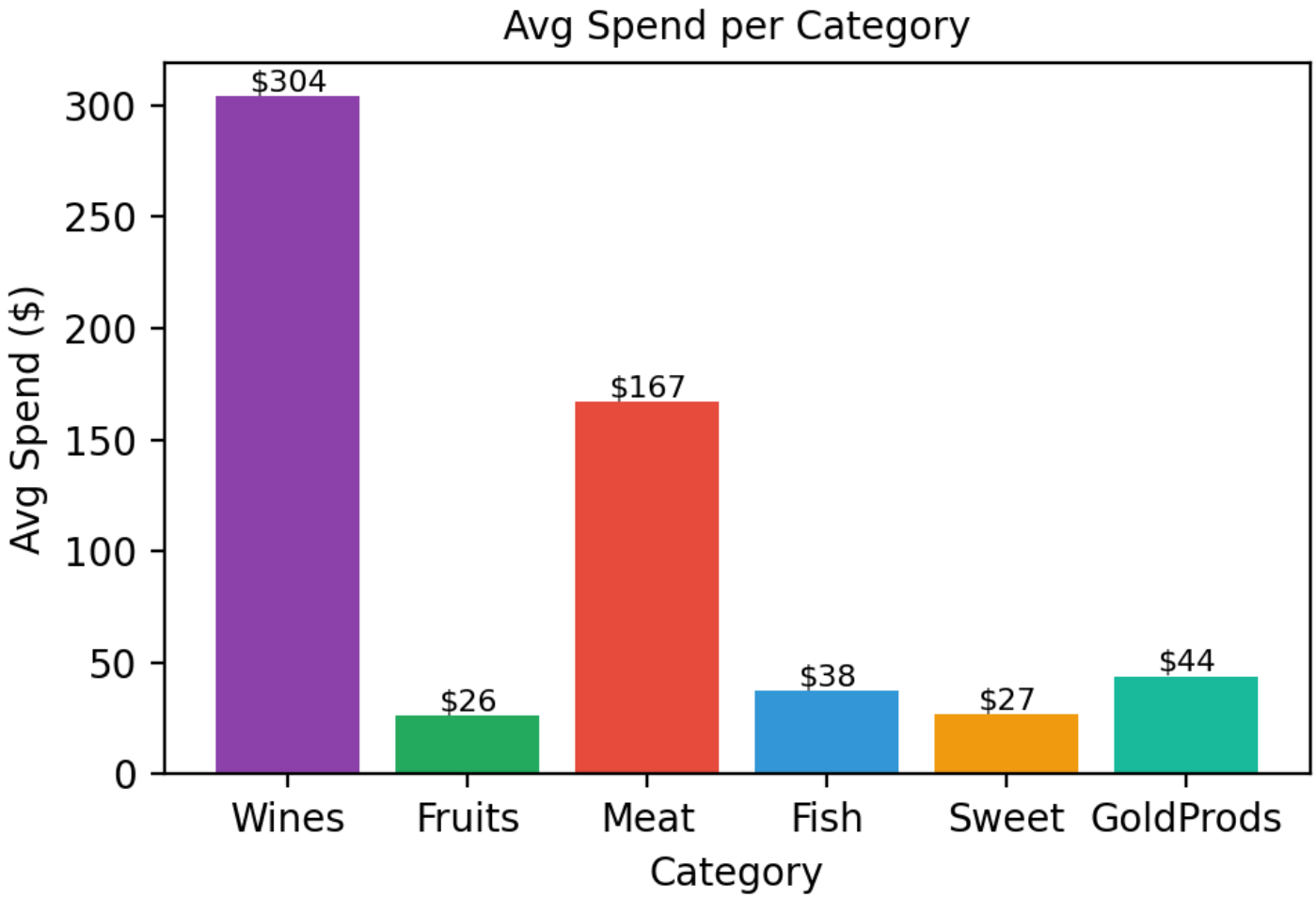
Segment	Count	Avg Income	Avg Spend	Accept Rate
Active Spenders	440	$69,924	$1,176	14.0%
Budget Shoppers	972	$34,501	$85	9.0%
Mid-tier Regulars	559	$56,814	$666	17.0%
Premium Customers	265	$75,871	$1,443	35.0%

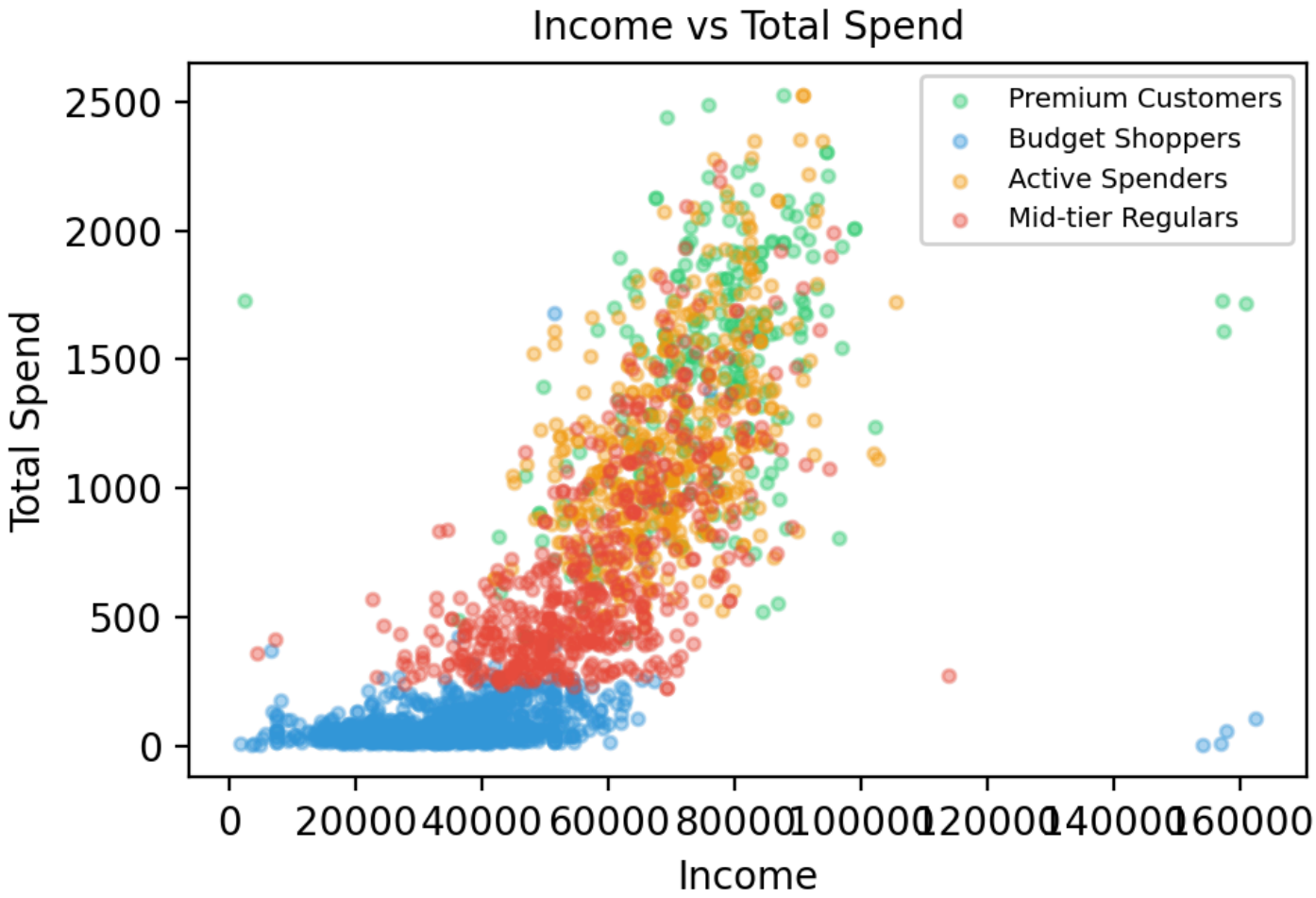
Cluster Profiles Comparison
Avg Income ($)
69924
34501
56814
75871
0
20000
40000
60000
Avg Spend ($)
1176
85
666
1443
0
500
1000
1500
Avg Purchases
24
7
18
21
0
10
20
Campaign Accept Rate
0
0
0
0
0.0
0.1
0.2
0.3
Active Spenders
Budget Shoppers
Mid-tier Regulars
Premium Customers

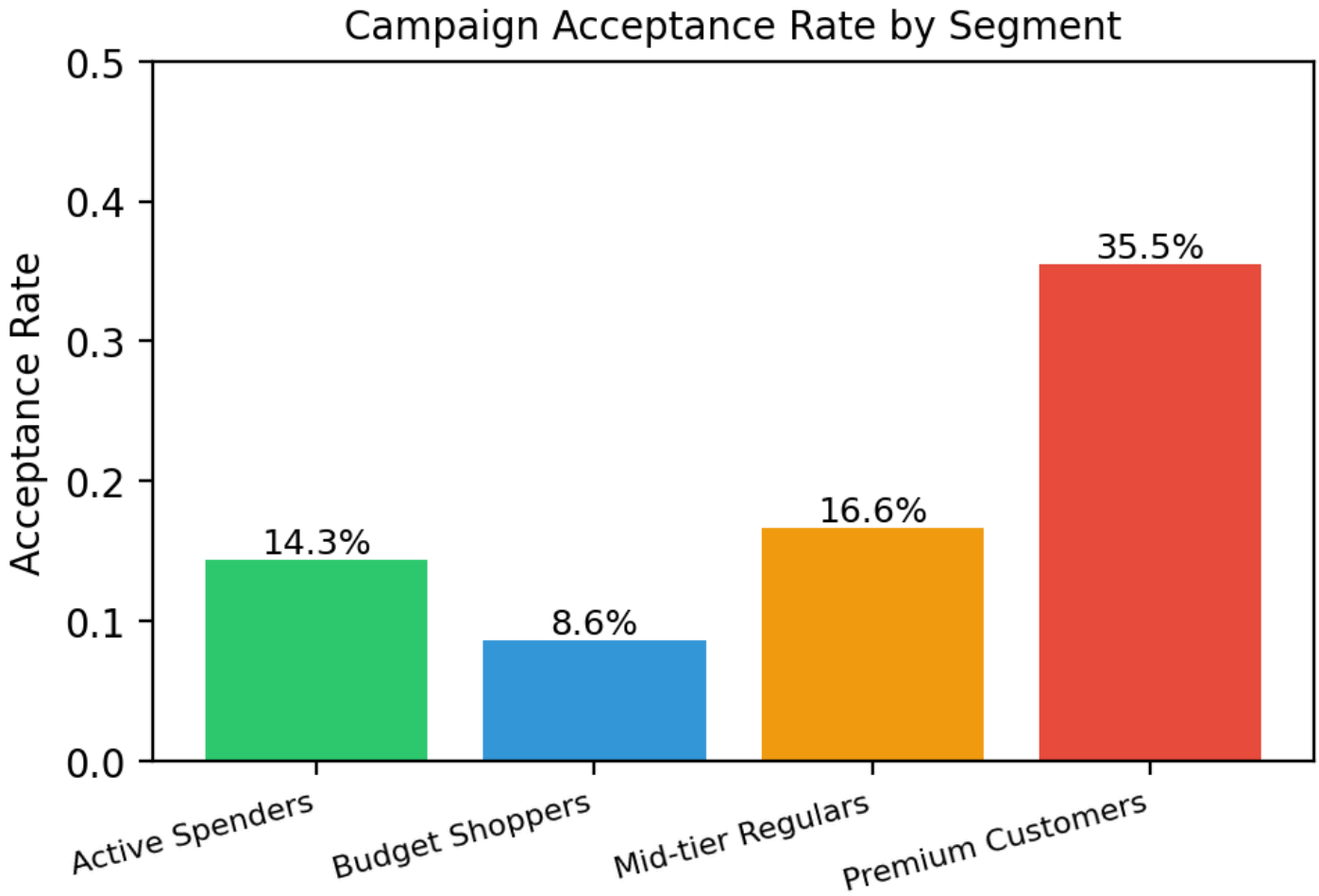

Segment Distribution
Premium Customers
11.9%
Active Spenders
19.7%
Mid-tier Regulars
25.0%
Budget Shoppers
43.5%

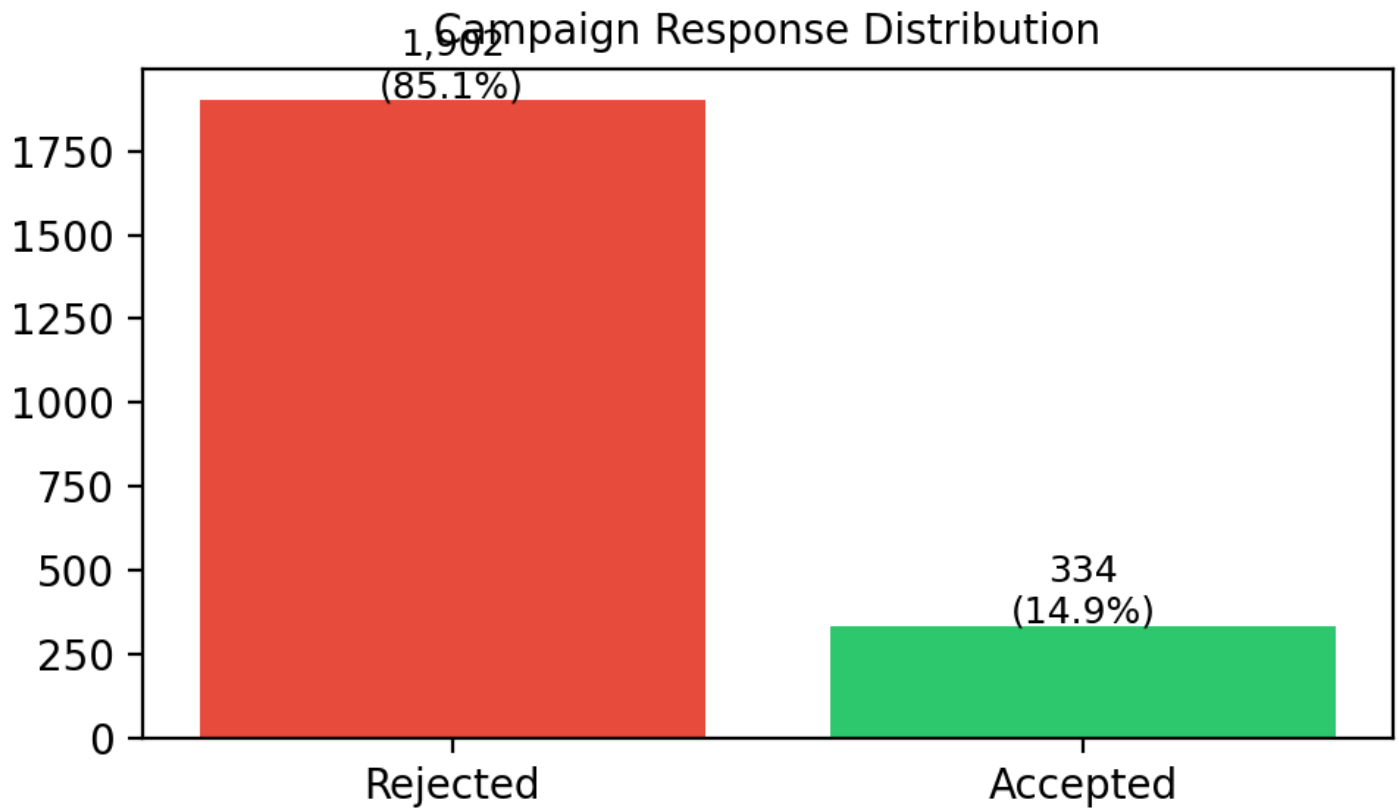

Income Distribution
Premium Customers
Budget Shoppers
Active Spenders
Mid-tier Regulars
250
200
150
100
50
0
0
20000
40000
60000
80000
100000
120000
140000
160000
Income

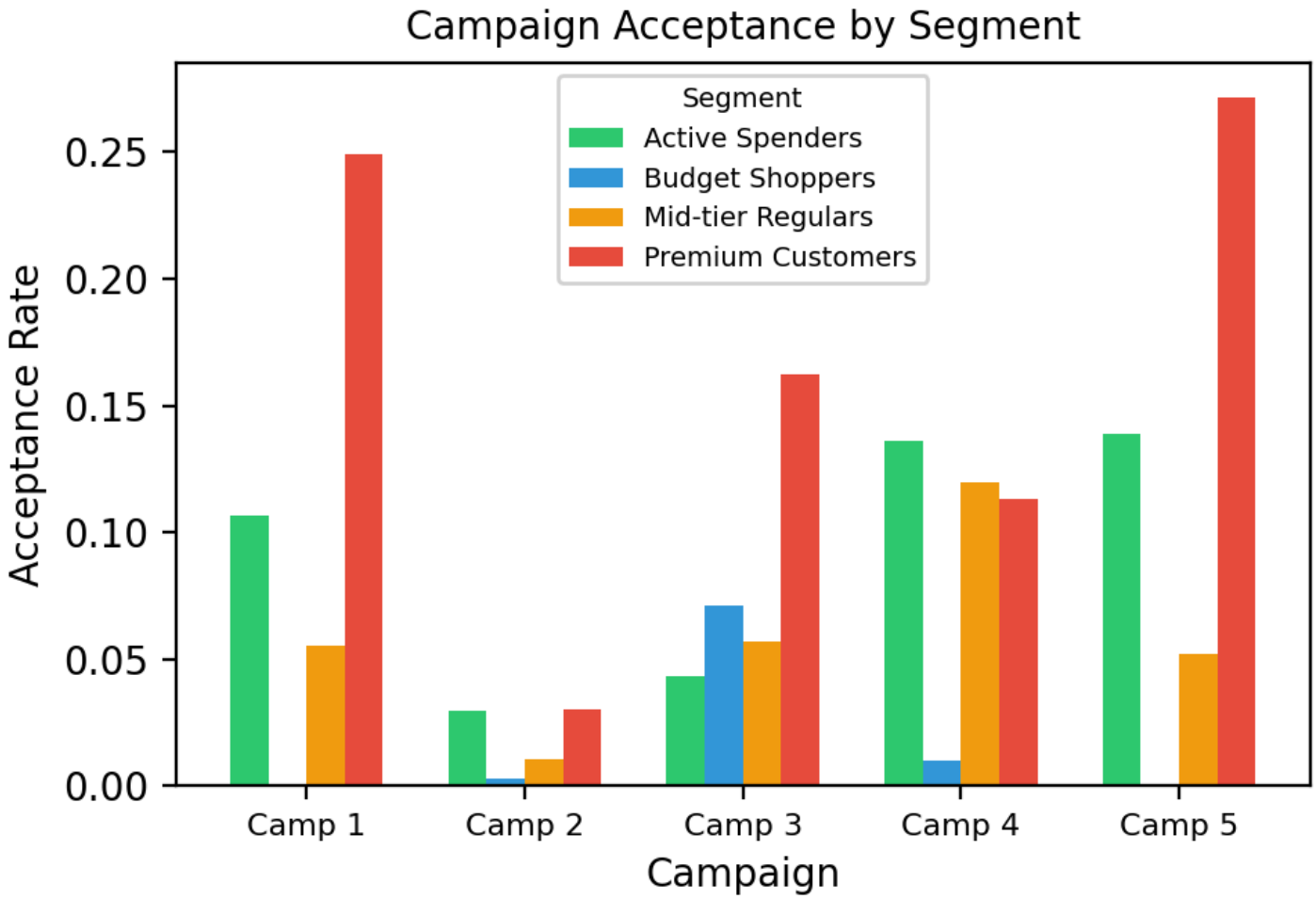

Avg Purchases by Channel
Avg Purchases
0
1
2
3
4
5
6
4.1
2.7
5.8
2.3
Web
Catalog
Store
Deals

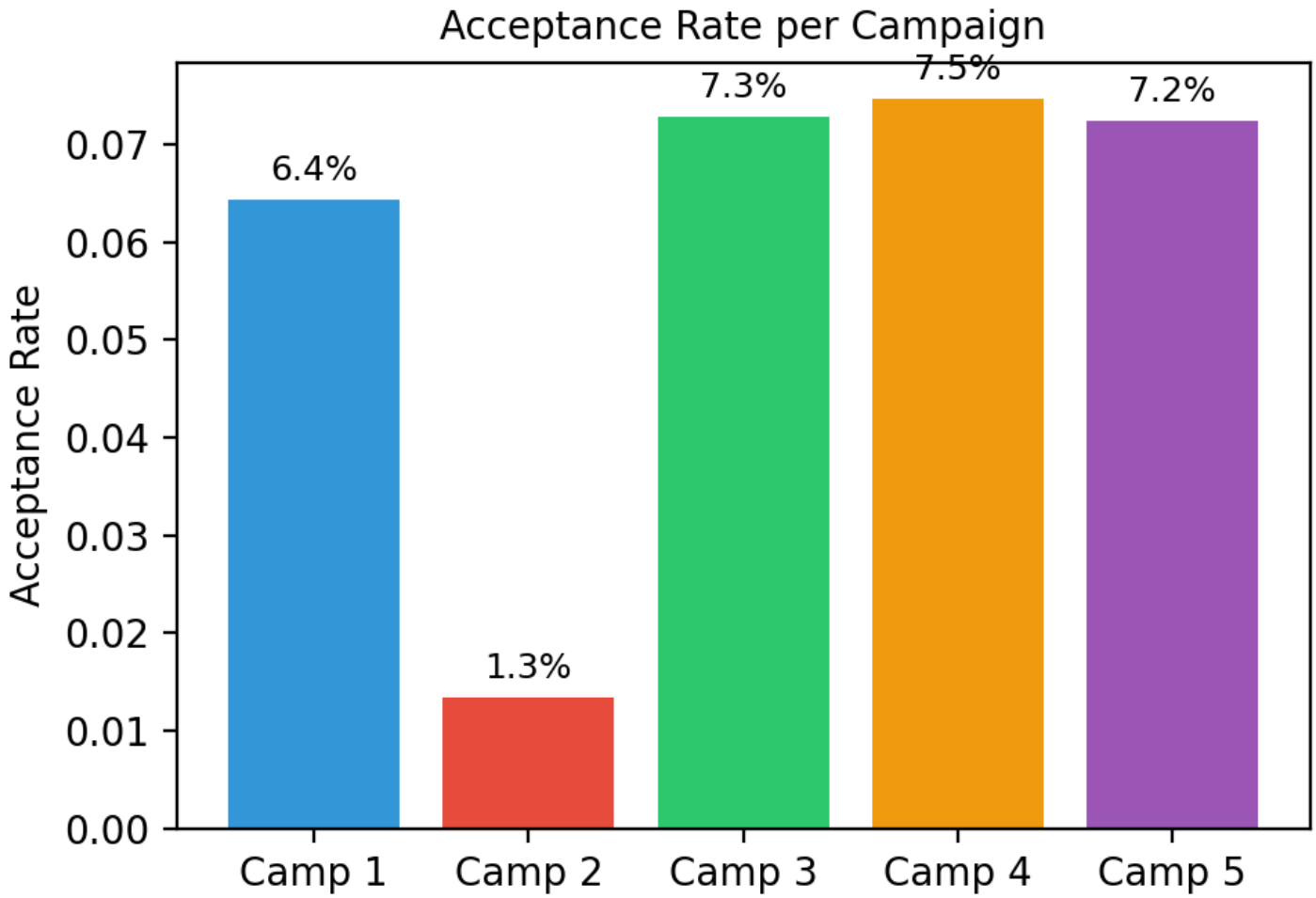

Avg Spend per Category
Avg Spend ($)
300
250
200
150
100
50
0
$304
$26
$167
$38
$27
$44
Wines
Fruits
Meat
Fish
Sweet
GoldProds
Category

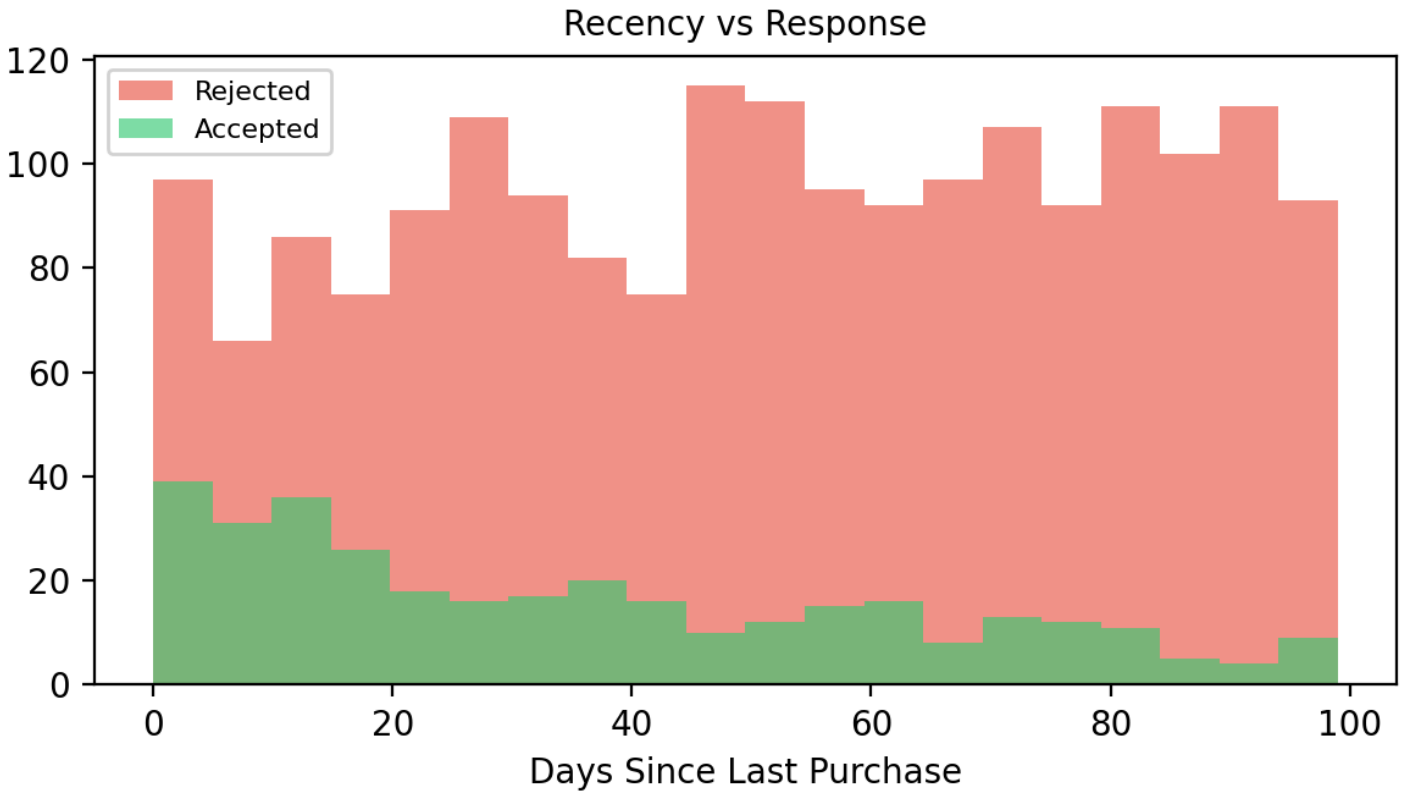

Income vs Total Spend
Premium Customers
Budget Shoppers
Active Spenders
Mid-tier Regulars
Total Spend
2500
2000
1500
1000
500
0
0
20000
40000
60000
80000
100000
120000
140000
160000
Income

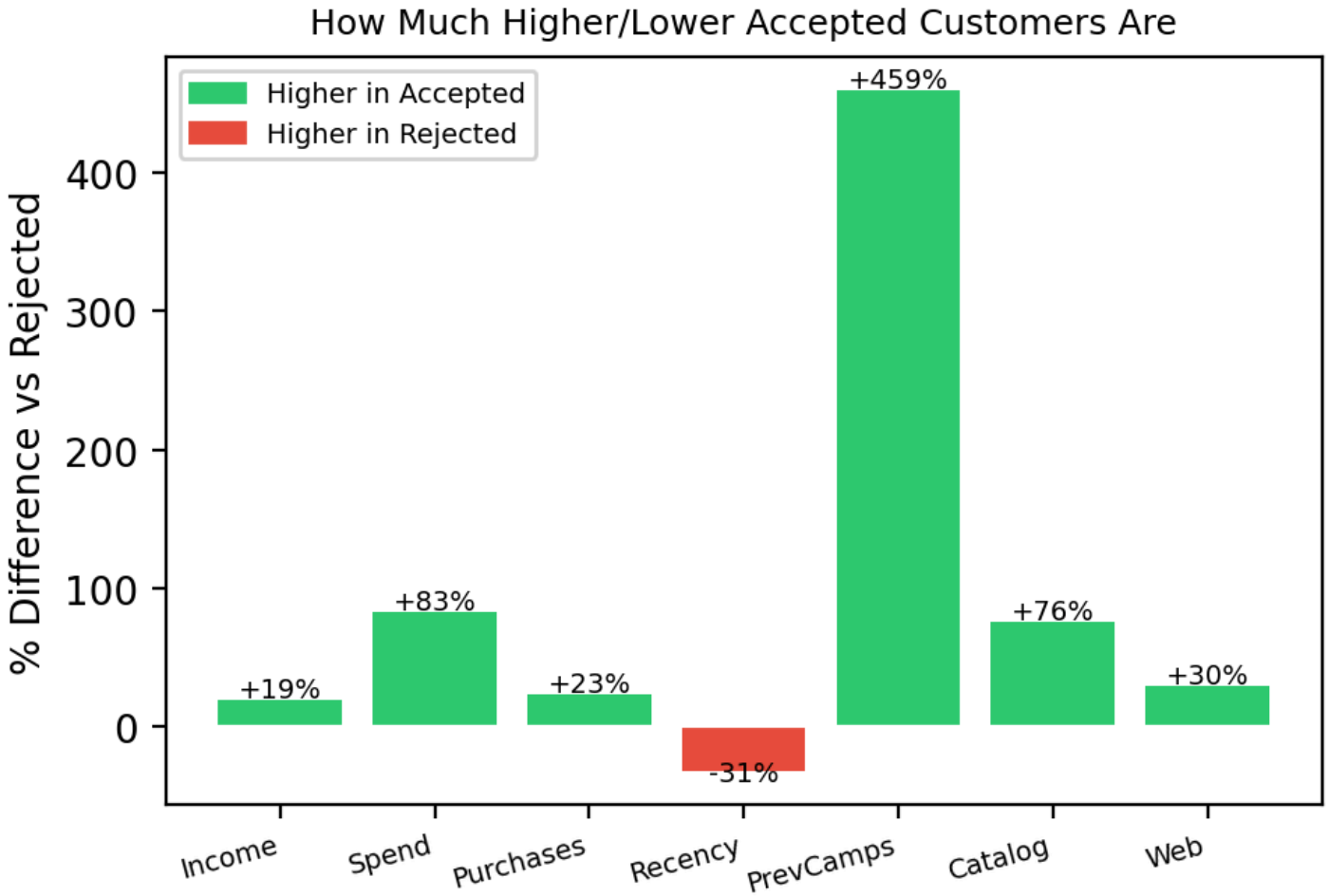

Campaign Acceptance Rate by Segment
Acceptance Rate
0.0
0.1
0.2
0.3
0.4
0.5
14.3%
8.6%
16.6%
35.5%
Active Spenders
Budget Shoppers
Mid-tier Regulars
Premium Customers


Campaign Response Distribution
0
250
500
750
1000
1250
1500
1750
1,902
(85.1%)
334
(14.9%)
Rejected
Accepted


Campaign Acceptance by Segment
Segment
Active Spenders
Budget Shoppers
Mid-tier Regulars
Premium Customers
Acceptance Rate
0.00
0.05
0.10
0.15
0.20
0.25
Camp 1
Camp 2
Camp 3
Camp 4
Camp 5
Campaign


Acceptance Rate per Campaign
Acceptance Rate
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
6.4%
1.3%
7.3%
7.5%
7.2%
Camp 1
Camp 2
Camp 3
Camp 4
Camp 5


Recency vs Response
Rejected
Accepted
0
20
40
60
80
100
120
0
20
40
60
80
100
Days Since Last Purchase


How Much Higher/Lower Accepted Customers Are
Higher in Accepted
Higher in Rejected
% Difference vs Rejected
400
300
200
100
0
+19%
+83%
+23%
-31%
+459%
+76%
+30%
Income
Spend
Purchases
Recency
PrevCamps
Catalog
Web